

Sri Lanka Institute of Information Technology

BSc (Hons) Information Technology
Specializing in Data Science



IT3021 – Data Warehousing and Business Intelligence
Year 03 Semester 01

Assignment 2

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1. Introduction

This report documents the end-to-end development of a Business Intelligence solution built upon the Superstore Retail Sales Data Warehouse, implemented in Assignment 1. The solution covers the full BI pipeline from data warehouse design through SSAS cube development, OLAP analysis in Microsoft Excel, and interactive reporting in Power BI.

The Superstore dataset represents a US-based retail company operating across four geographic regions — Central, East, South, and West — selling products across three main categories: Furniture, Office Supplies, and Technology. The dataset contains 9,994 order line items spanning five years (2014–2018), across 793 unique customers and 1,862 distinct products.

The key objectives of Assignment 2 are:

- Build and deploy an SSAS Multidimensional Cube using Superstore_DW as the data source.
- Define cube measures, connect all dimension tables, and implement hierarchies.
- Demonstrate all five OLAP operations — Roll-up, Drill-Down, Slice, Dice, and Pivot — using Excel PivotTables.
- Develop four Power BI reports covering Matrix groupings, cascading slicers, drill-down, and drill-through functionality.

2. Data Source for Assignment 2

2.1 Data Warehouse Overview

The data source for Assignment 2 is the Superstore_DW database, implemented and populated in Assignment 1. It is a star-schema data warehouse hosted on SQL Server and serves as the single analytical store for all reporting and cube operations in this assignment. The warehouse contains 9,994 sales transaction records spanning 2014–2018.

The data warehouse consolidates data from three heterogeneous source systems:

- Superstore_SourceDB (SQL Server) — Transactional order and order detail records.
- Customers_Source.xlsx (Excel) — Customer master data exported from a CRM system.
- Products_Source.xlsx (Excel) — Product catalogue and category hierarchy data.

The Superstore Data Warehouse employs a Star Schema with one central fact table and five dimension tables, all tables use surrogate keys and maintain referential integrity through foreign key relationships defined in the Data Source View.

- FactSales
 - records one row per order line item (one product in one order)
 - .Measures Sales, Profit, Quantity, Discount, and txn_process_time_hours. Accumulating timestamps accm_txn_create_time and accm_txn_complete_time track transaction lifecycle.
- DimCustomer (SCD Type 2)
 - CustomerID, CustomerName, and Segment.
 - StartDate and EndDate track historical Segment changes.
- DimProduct
 - ProductID, ProductName, SubCategory, and Category, supporting a three-level product hierarchy.
- DimLocation
 - City, State, PostalCode, Region, and Country, supporting a four-level geographic hierarchy.
- DimShipMode
 - ShipMode as a simple lookup dimension.
- DimDate
 - DateKey, Date, Day, Month, MonthName, Quarter, QuarterName, Year, DayOfWeek, DayName, and IsWeekend, supporting a four-level date hierarchy.

2.2 Star Schema Diagram

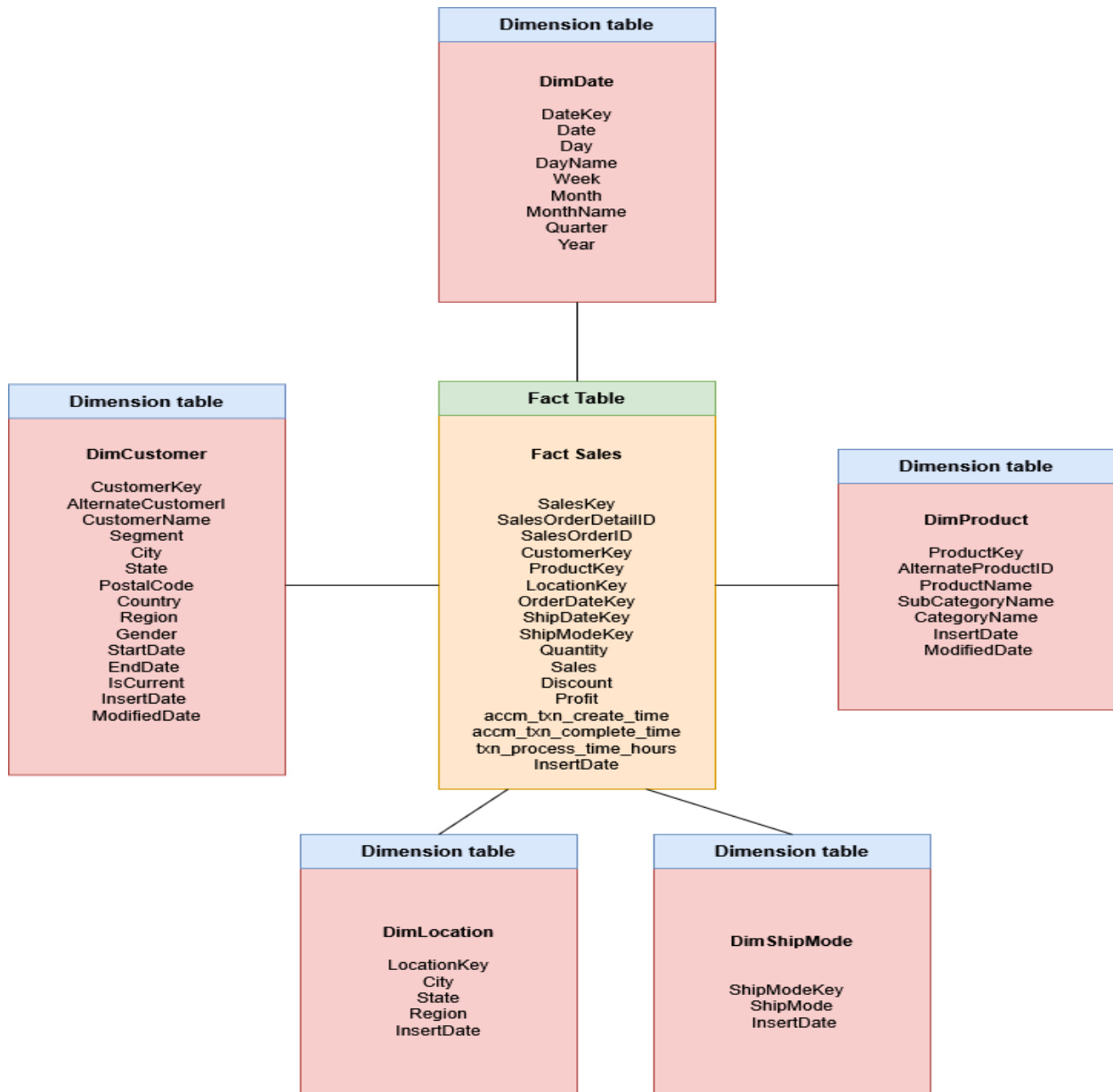


Figure: Star Schema of Superstore_DW

2.3 Database Instances

The three SQL Server databases used:

SQL Server – OLE DB Connections:

- Superstore_SourceDB — source OLTP database (Orders, OrderDetails)
- Superstore_Staging — staging database (5 staging tables)
- Superstore_DW — the data warehouse (fact + 5 dimensions)

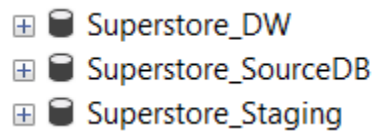


Figure: SSMS Object Explorer showing Superstore_SourceDB, Superstore_Staging, Superstore_DW

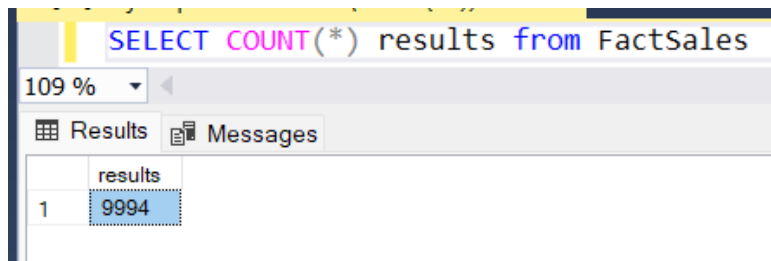


Figure: SSMS query: SELECT COUNT() FROM Superstore_DW.dbo.FactSales showing 9994*

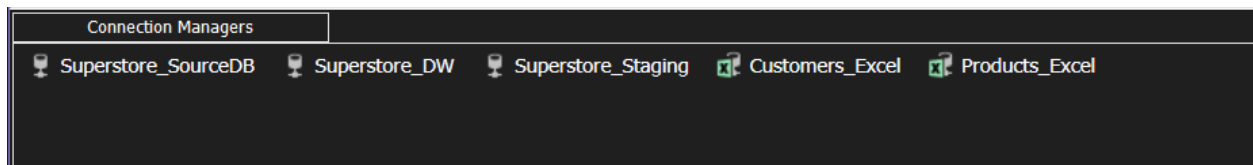


Figure : Connection Managers for all source types (3 OLE DB + 2 Excel).

3. SSAS Cube Implementation

3.1 Project Creation

The SSAS project was created using SQL Server Data Tools (SSDT) within Visual Studio 2022. The steps followed were:

1. Open Visual Studio 2022 as Administrator.
2. Navigate to File → New → Project.
3. Under Business Intelligence → Analysis Services, select Analysis Services Multidimensional and Data Mining Project.
4. Name the project Superstore_SSAS and set the solution directory.
5. Click OK to create the project.

Note: The project creates a solution structure containing Data Sources, Data Source Views, Cubes, and Dimensions folders in Solution Explorer.

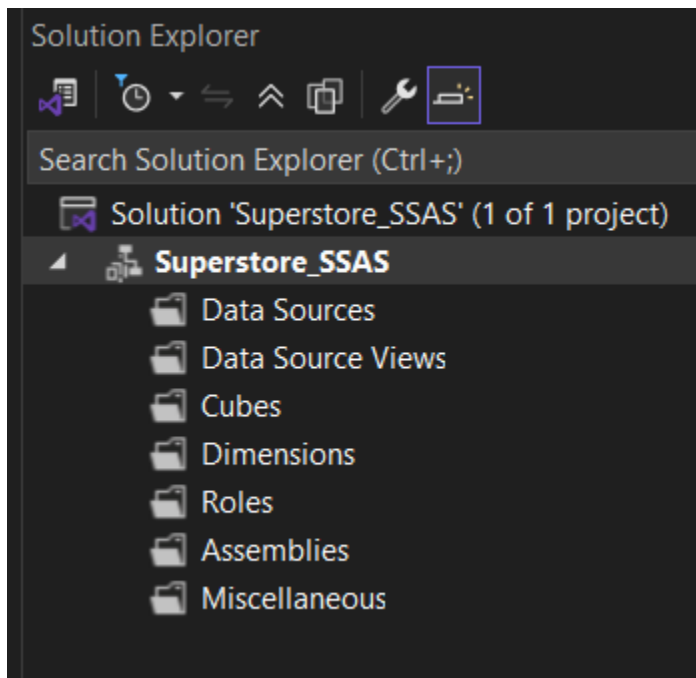


Figure: SSAS Project Structure in Solution Explorer showing Superstore_SSAS with Data Sources, Data Source Views, Cubes, and Dimensions folders

3.2 Data Source Definition

A data source connection was established to the Superstore_DW SQL Server database:

1. In Solution Explorer, right-click Data Sources → New Data Source.
2. In the Data Source Wizard, click New to create a new connection.
3. Select Microsoft OLE DB Provider for SQL Server.
4. Enter the server name (localhost\SQLEXPRESS) and select Superstore_DW database.
5. Set Impersonation to Use the service account.
6. Name the data source DS_SuperstoreDW and click Finish.

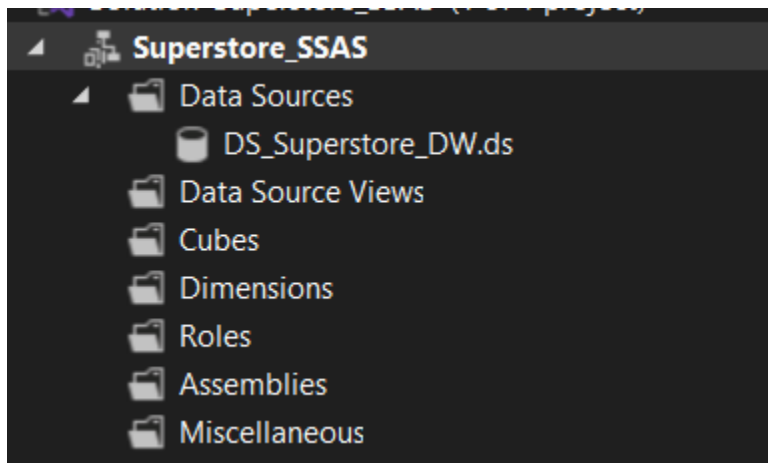


Figure: Data Source wizard showing DS_Superstore_DW created under Data Sources

3.3 Data Source View (DSV)

The Data Source View (DSV) provides a logical abstraction of the physical database schema used by the cube. Steps followed:

1. Right-click Data Source Views → New Data Source View.
2. Select DS_SuperstoreDW as the data source.
3. Add FactSales as the primary table, then click Add Related Tables to automatically pull in all five dimension tables.
4. Verify all foreign key relationships are correctly shown in the DSV diagram.
5. Name the DSV DSV_Superstore and save.

The DSV included: FactSales, DimDate, DimCustomer, DimProduct, DimLocation, and DimShipMode, with all relationships correctly defined.

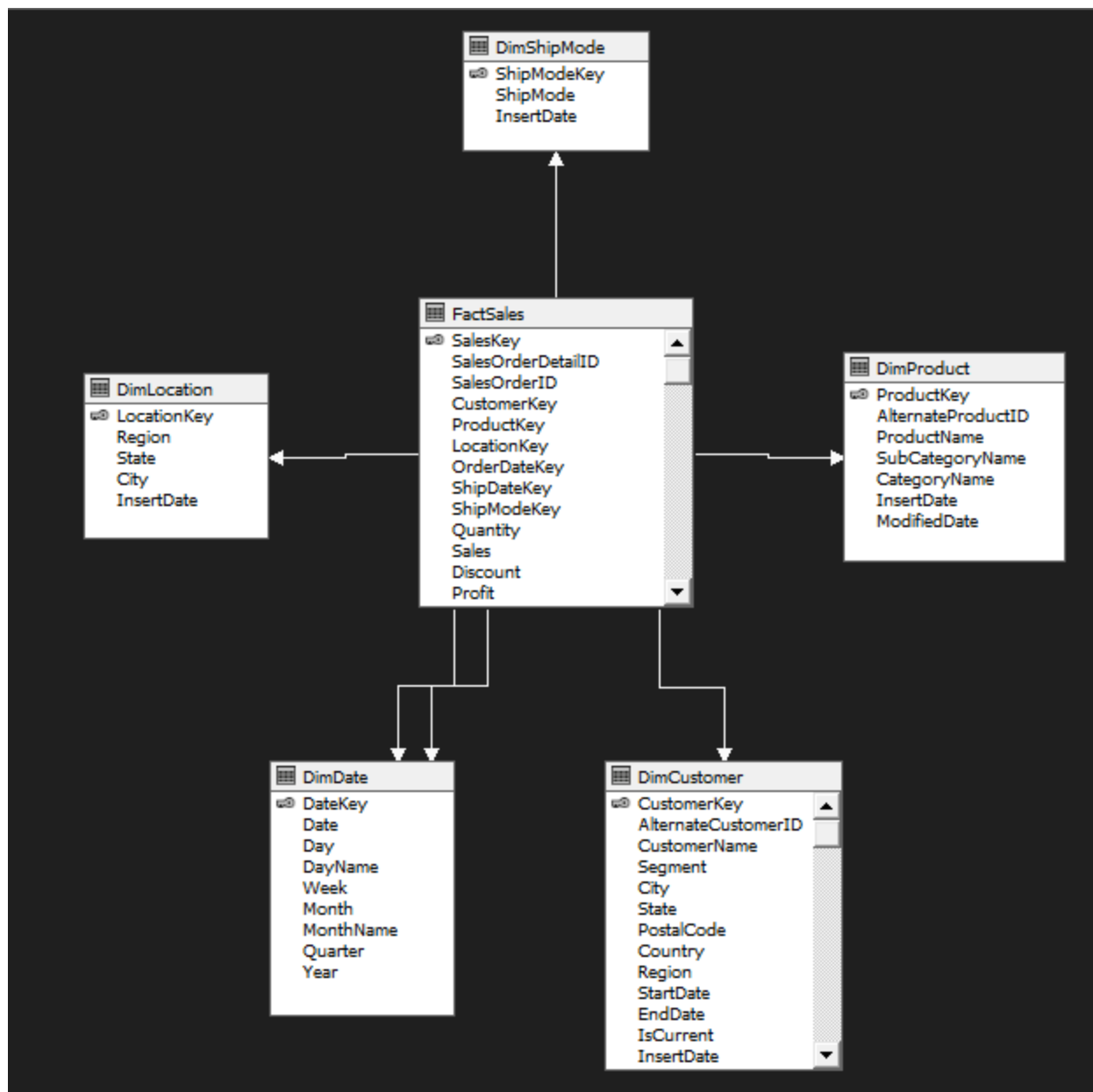


Figure: DSV design view showing FactSales connected to all 5 dimension tables with relationship lines

3.4 Cube Creation

The SSAS cube was created using the Cube Wizard:

1. Right-click Cubes → New Cube.
2. Select Use existing tables and click Next.
3. Select FactSales as the measure group table.
4. Select the following measures to include in the cube:
 - Sales
 - Quantity
 - Discount
 - Profit
5. Add all dimension tables: DimDate, DimCustomer, DimProduct, DimLocation, DimShipMode.
6. Name the cube Cube_Superstore and click Finish.

Two role-playing dimensions were created for DimDate, one for Order Date and one for Ship Date, to allow independent filtering of sales by order date versus ship date.

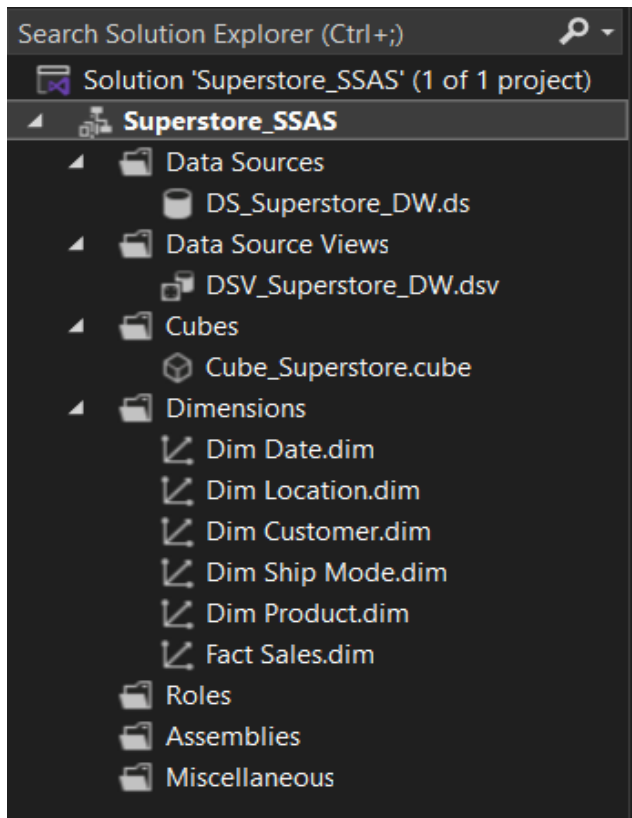


Figure : SSAS Project Structure in Solution Explorer, displaying the configured Data Source, Data Source View, Cube, Dimensions, and Role objects for the Superstore_SSAS solution.

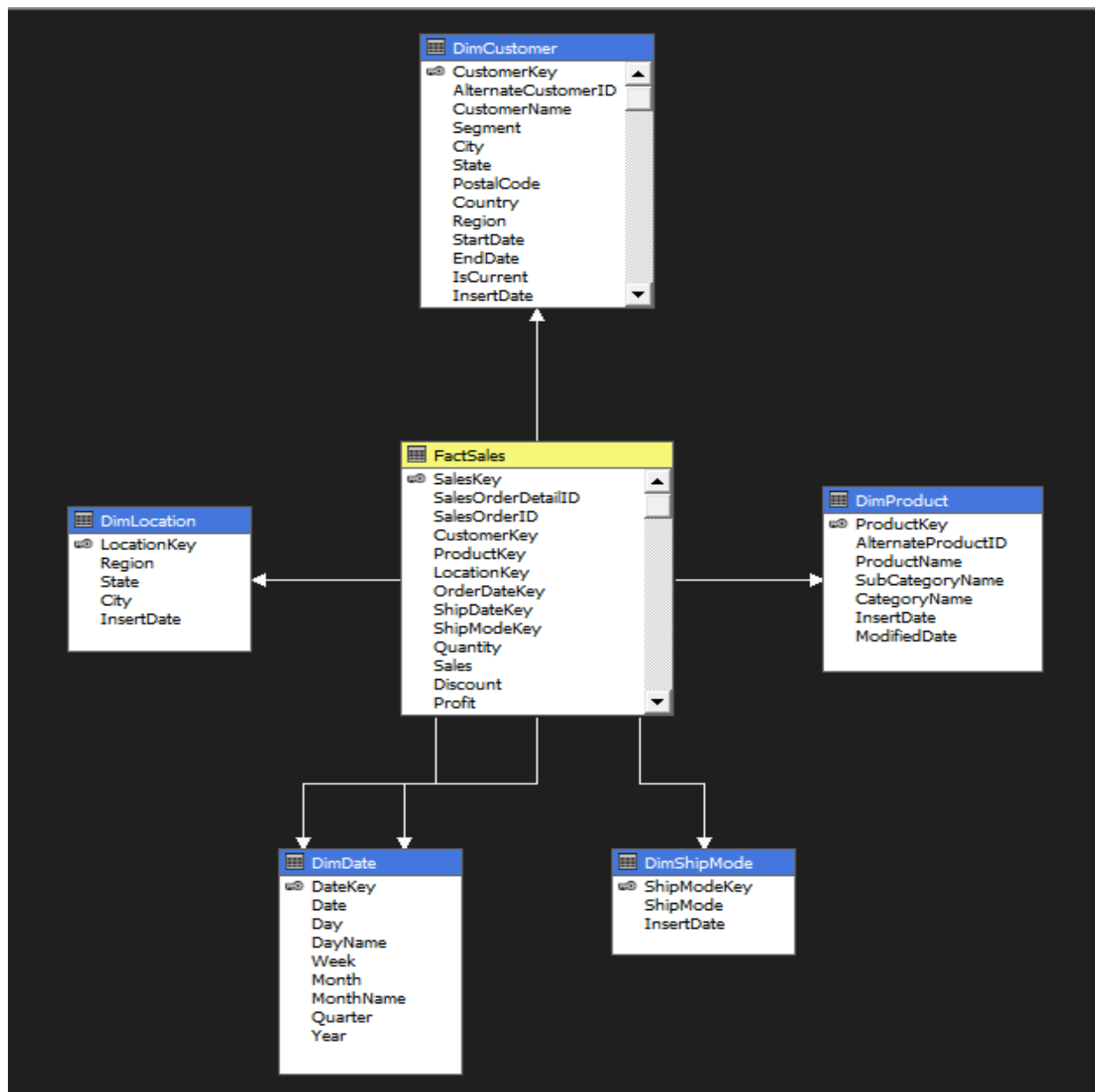


Figure: Cube design view showing FactSales measures and all 5 dimensions connected

3.5 Attribute Relationships and Hierarchies

3.5.1 Dim Date

Attribute Relationships

Attribute relationships were carefully configured in Dim Date to support the date hierarchy and prevent duplicate key errors during processing. The chain defined was:

Date Key → Date → Month → Quarter → Year

These relationships were created manually in the Attribute Relationships tab in SSDT by dragging and linking attributes accordingly.



DateHierarchy

A user-defined hierarchy named DateHierarchy was implemented with the following levels:

Year → Quarter → Month → Date

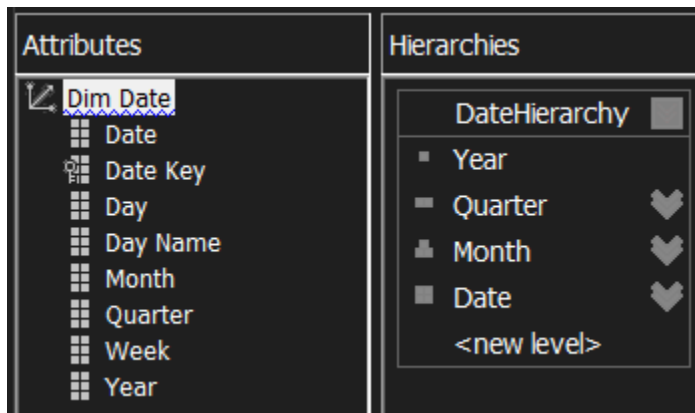


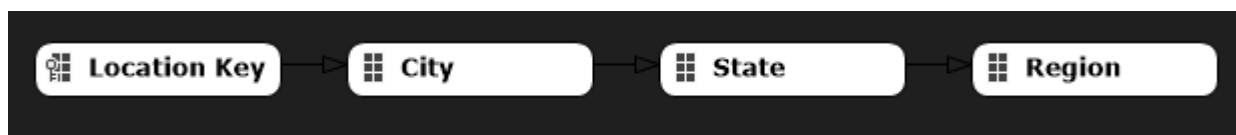
Figure : DateHierarchy in Dim Date — enables Year → Quarter → Month → Date drill-down

3.5.2 Dim Location

Attribute Relationships

Dim Location stores one row per unique city, meaning City is the most granular level. State and Region are parent levels. The attribute chain was configured as:

Location Key → City → State → Region



City used a composite key of State + City to uniquely identify cities that share names across different states (e.g., Arlington appears in both Texas and Virginia). State and Region each used their own single column as KeyColumns.

LocationHierarchy

A user-defined hierarchy named LocationHierarchy was implemented:

Region → State → City

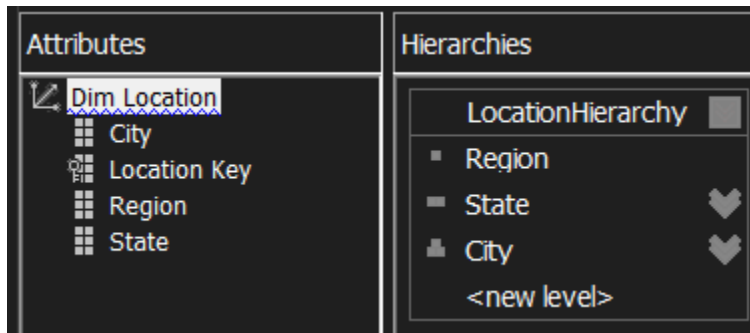


Figure : LocationHierarchy in Dim Location — enables Region → State → City drill-down

3.5.3 Dim Product

Attribute Relationships

The Dim Product dimension supports a three-level product hierarchy. Sub Category Name used a composite key of SubCategoryName + CategoryName since the same sub-category name may not be globally unique. The attribute chain was:

Product Key → Product Name → Sub Category Name → Category Name



ProductHierarchy

A user-defined hierarchy named ProductHierarchy was implemented:

Category Name → Sub Category Name → Product Name

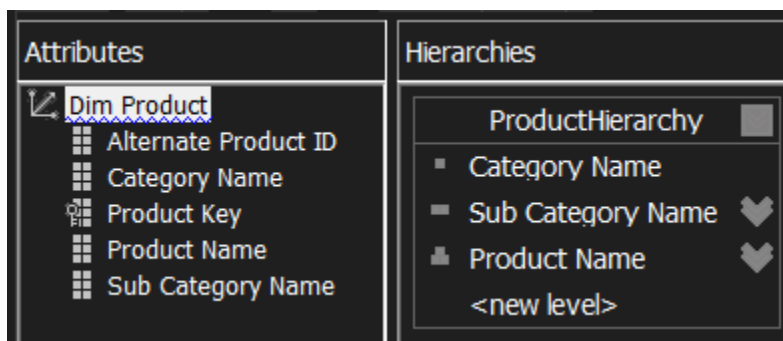


Figure : ProductHierarchy in Dim Product — enables Category → SubCategory → Product drill-down

3.6 Deployment and Processing

After completing the cube design, the project was deployed and processed:

1. Build the project: Build → Build Superstore_SSAS.
2. Deploy: Right-click the project in Solution Explorer → Deploy. The deployment target was localhost\SQLEXPRESS (Analysis Services instance).
3. Verify: The Output window showed 'Deploy: 1 succeeded, 0 failed, 0 skipped.'
4. Process the cube: Open SSMS → Connect to Analysis Services → Right-click Superstore_SSAS → Process Database → Process Full.
5. Confirmed all dimensions and the fact table processed successfully with zero errors.

During the processing stage, several duplicate key issues were identified and resolved by correctly configuring composite keys in Dim Date (Month, Quarter), Dim Location (City), and Dim Product (Sub Category Name). The final processing run completed all dimensions and the FactSales measure group with no errors.

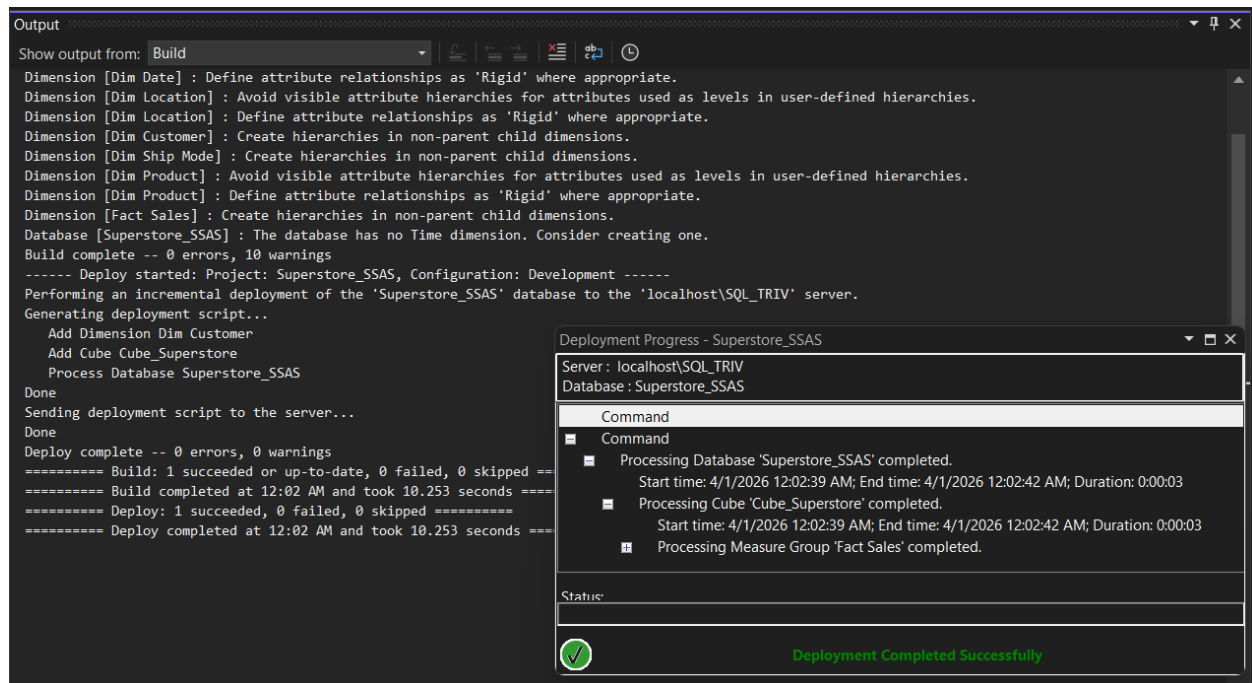


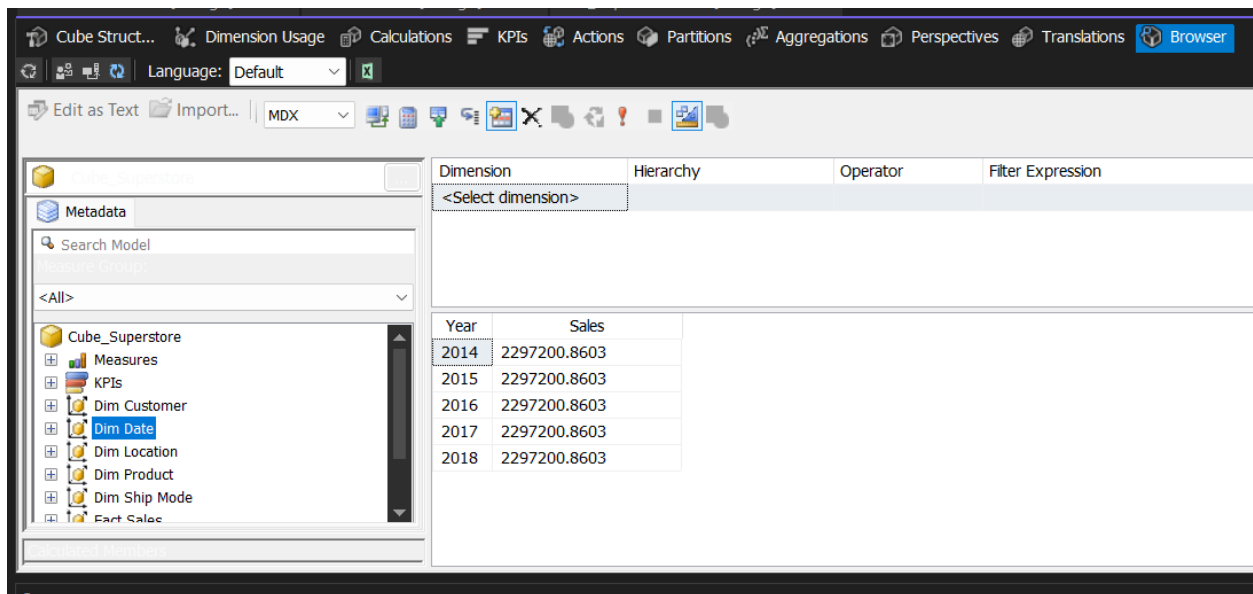
Figure: Deployment successful dialog showing 'Deployment Completed Successfully' with no errors

3.7 Cube Browsing and Validation

The deployed cube was validated using two methods:

3.7.1 Browser Tab in Visual Studio

1. Open Cube_Superstore.cube in VS 2022 → click the Browser tab.
2. From the left Metadata panel, expand Measures → Fact Sales → drag Sales to the data area.
3. Expand Dim Date → DateHierarchy → drag Year to the Rows area.
4. Click 'Click to execute the query'.
5. The browser displayed Sales totals aggregated by Year, confirming the cube is processing data correctly.



Dimension	Hierarchy	Operator	Filter Expression
<Select dimension>			

Year	Sales
2014	2297200.8603
2015	2297200.8603
2016	2297200.8603
2017	2297200.8603
2018	2297200.8603

Figure : Cube browser validation showing Sales by Year — confirms successful cube deployment and processing

3.7.2 SSMS Analysis Services Browser

1. In SSMS, connect using Analysis Services as the server type (localhost).
2. Expand Databases → Superstore_SSAS → Cubes.
3. Right-click Cube_Superstore → Browse.
4. Drag Sales to the data area and Year to rows, confirming the cube is accessible and returns data from SSMS.

SQLQuery1.sql - L...6EAMPD\USER (63))*

Language: Default

MDX

Cube_Superstore

Metadata

Search Model

Measure Group:

<All>

Cube_Superstore

- Measures
- KPIs
- Dim Customer
- Dim Date
- Dim Location
- Dim Product
- Dim Ship Mode
- Fact Sales
- Order Date
- Ship Date

Dimension

<Select dimension>

Hierarchy

Year	Sales
2014	2297200.8603
2015	2297200.8603
2016	2297200.8603
2017	2297200.8603
2018	2297200.8603

These two validation methods confirmed that the cube was successfully deployed, dimensions were correctly linked to FactSales, hierarchies were navigable, and measures were accurately aggregating data from the underlying Superstore data warehouse.

4. Demonstration of OLAP Operations Using Excel

This section demonstrates the OLAP capabilities of the Superstore SSAS cube by connecting an Excel workbook and performing standard OLAP operations including Roll-up, Drill-Down, Slice, Dice, and Pivot.

Each operation is illustrated through corresponding PivotTable views and screenshots

4.1 Connecting Excel to the Cube

Microsoft Excel was connected to the deployed SSAS cube to enable PivotTable-based OLAP analysis:

1. Open Microsoft Excel → click the Data tab.
2. Click Get Data → From Database → From Analysis Services.
3. Enter server name: localhost (or localhost\SQLEXPRESS as applicable).
4. Select database: Superstore_SSAS.
5. Select cube: Cube_Superstore.
6. Click Insert PivotTable to create a new PivotTable connected to the live cube.

The PivotTable Field List appeared on the right, showing all cube dimensions (Dim Date, Dim Customer, Dim Product, Dim Location, Dim Ship Mode) and the Fact Sales measure group, including the Sales, Quantity, Discount, and Profit measures.

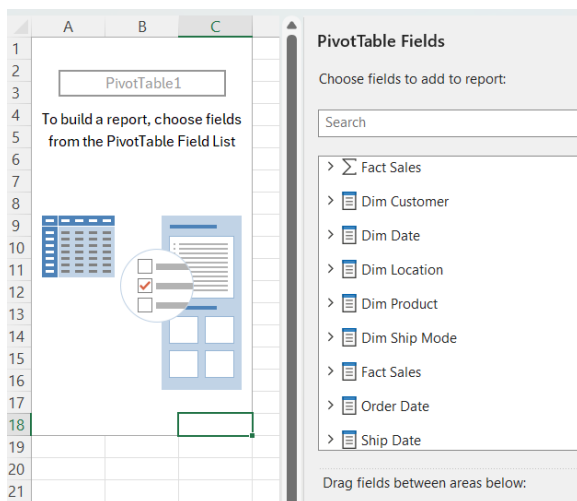


Figure: Excel PivotTable Field List showing Superstore SSAS cube dimensions and measures

4.2 Roll-Up and Drill-Down Operations

Roll-up and Drill-Down were demonstrated using the DateHierarchy from the Order Date dimension.

PivotTable Configuration:

- Rows: Order Date → DateHierarchy
- Values: Sales (from Fact Sales → Measures)

Roll-Up:

The PivotTable initially displays data at the Year level (most rolled-up view), showing Sales totals aggregated across all quarters, months, and days for each year. This represents the highest level of aggregation in the DateHierarchy — data is rolled up from individual order dates all the way to annual totals.

To demonstrate roll-up explicitly: first expanded all levels to show month-level detail, then clicked the (–) collapse button next to each month to roll up to Quarter, and again to roll up to Year.

The screenshot shows a PivotTable in Excel with the following data:

Row Labels	Sales
2014	2297200.86
2015	2297200.86
2016	2297200.86
2017	2297200.86
2018	2297200.86
Grand Total	2297200.86

The PivotTable Fields task pane on the right is configured as follows:

- Choose fields to add to report:** Search bar.
- Fact Sales:**
 - ☐ Discount
 - ☐ Fact Sales Count
 - ☐ Profit
 - ☐ Quantity
 - ☒ Sales
 - ☐ Sales Order Detail ID
- Drag fields between areas below:**
 - Filters:** (Empty)
 - Columns:** (Empty)
 - Rows:** Dim Date.DateHierarchy
 - Values:** Sales

Figure: Roll-Up: PivotTable showing Sales by YEAR only (highest level of DateHierarchy) — all years collapsed

Drill-Down:

To demonstrate Drill-Down, the (+) expand button was clicked next to 2017, revealing Quarter-level data. Expanding further revealed Month-level sales data. This shows how the DateHierarchy enables progressive navigation from Year → Quarter → Month → Date.

	A	B	C
1	Row Labels	Sales	
2	+ 2014	2297200.86	
3	+ 2015	2297200.86	
4	+ 2016	2297200.86	
5	- 2017		
6	+ 1	2297200.86	
7	+ 2	2297200.86	
8	+ 3	2297200.86	
9	+ 4	2297200.86	
10	+ 2018	2297200.86	
11	Grand Total	2297200.86	
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			

PivotTable Fields

Choose fields to add to report:

Search

Fact Sales

- ☐ Discount
- ☐ Fact Sales Count
- ☐ Profit
- ☐ Quantity
- ☒ Sales
- ☐ Sales Order Detail ID

Drag fields between areas below:

Filters	Columns
Rows	Values
Dim Date.DateHierarc...	Sales

Figure: Drill-Down — PivotTable expanded to Quarter level within 2017, showing progressive detail

4.3 Slice Operation

A Slice operation reduces the data cube by fixing one dimension to a single value, effectively viewing a single 'slice' of the cube.

PivotTable Configuration:

- Filters: Dim Location → Region = West
- Rows: Dim Product → ProductHierarchy (Category → Sub Category)
- Values: Sales

By setting the Region filter to 'West', the PivotTable displays only sales data for the West region, effectively slicing the entire cube on a single dimension value. The resulting table shows product category and sub-category sales breakdown exclusively for the West region, eliminating all Central, East, and South data from the view.

	A	B	C
1	LocationHierarchy	West	
2			
3	Row Labels	Sales	
4	Furniture	279274.9733	
5	Office Supplies	240075.064	
6	Technology	245284.408	
7	Grand Total	764634.4453	
8			
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22			
23			
24			

PivotTable Fields

Choose fields to add to report:

Search

Fact Sales

- ☐ Discount
- ☐ Fact Sales Count
- ☐ Profit
- ☐ Quantity
- ☒ Sales
- ☐ Sales Order Detail ID

Drag fields between areas below:

Filters	Columns
LocationHierarchy	
Rows	Values
ProductHierarchy	Sales

Figure : Slice Operation — PivotTable filtered to Region = West, showing product hierarchy sales for West region only

4.4 Dice Operation

A Dice operation applies multiple dimension filters simultaneously, creating a sub-cube from the intersection of selected values across two or more dimensions.

PivotTable Configuration:

- Filter 1: Dim Location → Region = West
- Filter 2: Order Date → Year = 2017
- Rows: Dim Product → ProductHierarchy
- Values: Sales

Two filters were applied simultaneously: Region = West AND Year = 2017. This dice operation restricts the data to the specific sub-cube representing West region sales in the year 2017 only, broken down by product hierarchy. The values in the PivotTable changed to reflect this specific two-dimensional filter intersection, demonstrating how Dice narrows the cube to a targeted analytical subset.

	A	B	C	D
1	LocationHierarchy	West		
2	Order Date.DateHierarchy	2017		
3				
4	Row Labels	Sales		
5	Furniture	85816.5712		
6	Office Supplies	83501.507		
7	Technology	70917.097		
8	Grand Total	240235.1752		
9				
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22				
23				
24				

PivotTable Fields
Choose fields to add to report:
Search

Fact Sales
☐ Discount
☐ Fact Sales Count
☐ Profit
☐ Quantity
☒ **Sales**
☐ Sales Order Detail ID

Drag fields between areas below:

Filters
LocationHierarchy
Order Date.DateHiera...

Columns

Rows
ProductHierarchy

Values
Sales

Figure : Dice Operation — PivotTable with two active filters (Region = West AND Year = 2017) showing sales by product category

4.5 Pivot Operation

A Pivot operation rotates the data view by moving dimensions between the Rows and Columns areas, changing the orientation of analysis without altering the underlying data.

Before Pivot:

- Rows: Dim Product → ProductHierarchy (Category → Sub Category)
- Columns: Dim Location → LocationHierarchy (Region → State → City)
- Values: Sales

After Pivot:

- Rows: Dim Location → LocationHierarchy (moved from Columns to Rows)
- Columns: Dim Product → ProductHierarchy (moved from Rows to Columns)
- Values: Sales

By moving ProductHierarchy from the Rows area to the Columns area and moving LocationHierarchy from Columns to Rows, the PivotTable orientation was rotated. This changed the analysis from a vertical product view with horizontal geographic columns to a vertical geographic view with horizontal product columns, enabling a different perspective on the same Sales data without changing any underlying values.

	A	B	C	D	E	F
1	Sales	Column Labels				
2	Row Labels	Central	East	South	West	Grand Total
3	Furniture	176419.4072	172460.7085	113844.7063	279274.9733	741999.7953
4	Office Supplies	157627.948	193251.332	128092.688	240075.064	719047.032
5	Technology	184752.777	246022.259	160094.589	245284.408	836154.033
6	Grand Total	518800.1322	611734.2995	402031.9833	764634.4453	2297200.86
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PivotTable Fields
Choose fields to add to report:
Search
Fact Sales
☐ Discount
☐ Fact Sales Count
☐ Profit
☐ Quantity
☒ Sales
☐ Sales Order Detail ID
Drag fields between areas below:
Filters
Columns
LocationHierarchy
Rows
ProductHierarchy
Values
Sales

Figure: Pivot BEFORE: ProductHierarchy in Rows, LocationHierarchy in Columns, Sales values

	A	B	C	D	E	F
1	Sales	Column Labels				
2	Row Labels	Furniture	Office Supplies	Technology	Grand Total	
3	Central	176419.4072	157627.948	184752.777	518800.1322	
4	East	172460.7085	193251.332	246022.259	611734.2995	
5	South	113844.7063	128092.688	160094.589	402031.9833	
6	West	279274.9733	240075.064	245284.408	764634.4453	
7	Grand Total	741999.7953	719047.032	836154.033	2297200.86	
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PivotTable Fields
Choose fields to add to report:
Search
Fact Sales
☐ Discount
☐ Fact Sales Count
☐ Profit
☐ Quantity
☒ Sales
☐ Sales Order Detail ID
Drag fields between areas below:
Filters
Columns
ProductHierarchy
Rows
LocationHierarchy
Values
Sales

Figure: Pivot AFTER: LocationHierarchy in Rows, ProductHierarchy in Columns — same data, rotated perspective

5. Power BI Reports

5.1 Data Preparation

All four Power BI reports were developed using a live connection to the deployed Superstore_SSAS cube. A live connection ensures that all data aggregations and hierarchies are handled by the cube engine rather than Power BI, preserving the integrity of the OLAP model.

Connection was established in Power BI Desktop by:

1. Open Power BI Desktop → Get Data → Analysis Services.
2. Enter server: localhost (or the applicable Analysis Services instance name).
3. Select Superstore_SSAS database.
4. Select Cube_Superstore.
5. Choose Connect live to maintain a live cube connection.

The following measures and dimensions were used across all reports:

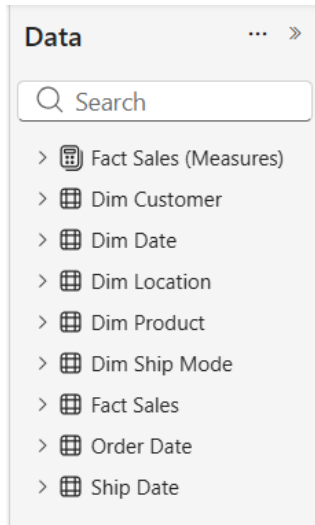
- Measures: Sales, Quantity, Profit, Discount (from Fact Sales)
- Dimensions: Dim Date (DateHierarchy), Dim Product (ProductHierarchy), Dim Location (LocationHierarchy), Dim Customer, Dim Ship Mode

5.2 Data Modeling

Since the connection was live, no additional relationships were created in Power BI Desktop. All hierarchies (DateHierarchy, LocationHierarchy, ProductHierarchy) were sourced directly from the cube and used as-is in all visuals. No calculated columns or new tables were added — all aggregations were delegated to the cube.

5.3 Use of DAX

No custom DAX expressions were created for these reports. All measures were sourced directly from the SSAS cube, which already contains correctly aggregated additive measures (Sales, Quantity, Profit, Discount). Power BI's native aggregation functions (Sum, Average) were applied at the visual level where needed, consistent with the cube model.



Power BI Desktop connected to SSAS cube — Fields panel showing Dim Customer, Dim Date, Dim Location, Dim Product, Dim Ship Mode, Fact Sales

5.4 Visual Design

All reports were designed with a consistent professional visual theme:

- Color scheme: Navy blue (#1F3864) for headers, medium blue (#2E75B6) for accents, light grey (#F5F8FC) for backgrounds.
- Each report is on a separate page with a clear title.
- All visuals include axis labels, data labels, and legends where applicable.
- Interactive elements (slicers, drill-down arrows, drill-through context menu) are clearly accessible.

5.5 Report 1: Matrix Visual — Sales and Profit by Product Category and Year

The first Power BI report presents a Matrix visual designed to display detailed tabular sales data with hierarchical row groupings and year-based column groupings. The primary objective of this report is to allow users to analyse Sales and Profit performance across product categories and sub-categories over the four-year period from 2014 to 2018.

The Matrix visual was configured with the ProductHierarchy defined in DimProduct as the row grouping, providing three levels: Category Name at the top level, Sub Category Name at the second level, and Product Name at the most granular level. These levels were added sequentially into the Rows field of the Matrix. The Year field from DimDate was placed in the Columns field, creating a cross-tabulation view showing 2014, 2015, 2016, 2017 and 2018 as column headers.

Sales and Profit from FactSales were added to the Values section of the Matrix. Row subtotals and column grand totals were enabled, providing aggregate figures at each level of the hierarchy. A Region slicer was added to the report canvas, allowing users to filter the entire Matrix to a specific geographic region.

The Matrix supports Expand and Collapse functionality. At the Category level, users see aggregated Sales and Profit for Furniture, Office Supplies, and Technology. Clicking the expand arrow reveals Sub-Category level detail, and expanding further reveals individual Product Name level figures. This hierarchical navigation enables both roll-up summarisation and drill-down detail analysis within a single visual.

The report revealed that Technology generated the highest total Sales across all four years, while Office Supplies showed the highest volume of sub-categories. Profit margins varied significantly across sub-categories, with Tables in Furniture showing negative profit due to high discount rates.

Sales and Profit by Product Category and Years

Year	2014		2015		2016		2017		2018		Total	
Category Name	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit
□ Furniture	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27
□ Office Supplies	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80
□ Technology	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95
Total	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02

Figure: Report 1 — Matrix visual at Category level showing Sales and Profit for Furniture, Office Supplies, and Technology across years 2014 to 2018 with conditional. formatting (red=low, green=high)

Sales and Profit by Product Category and Years

Year	2014		2015		2016		2017		2018		Total	
Category Name	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit
▣ Furniture	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27	741,999.80	18,451.27
▣ Tables	206,965.53	-17,725.48	206,965.53	-17,725.48	206,965.53	-17,725.48	206,965.53	-17,725.48	206,965.53	-17,725.48	206,965.53	-17,725.48
▣ Bookcases	114,880.00	-3,472.56	114,880.00	-3,472.56	114,880.00	-3,472.56	114,880.00	-3,472.56	114,880.00	-3,472.56	114,880.00	-3,472.56
▣ Furnishings	91,705.16	13,059.14	91,705.16	13,059.14	91,705.16	13,059.14	91,705.16	13,059.14	91,705.16	13,059.14	91,705.16	13,059.14
▣ Chairs	328,449.10	26,590.17	328,449.10	26,590.17	328,449.10	26,590.17	328,449.10	26,590.17	328,449.10	26,590.17	328,449.10	26,590.17
▣ Office Supplies	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80	719,047.03	122,490.80
▣ Supplies	46,673.54	-1,189.10	46,673.54	-1,189.10	46,673.54	-1,189.10	46,673.54	-1,189.10	46,673.54	-1,189.10	46,673.54	-1,189.10
▣ Fasteners	3,024.28	949.52	3,024.28	949.52	3,024.28	949.52	3,024.28	949.52	3,024.28	949.52	3,024.28	949.52
▣ Labels	12,486.31	5,546.25	12,486.31	5,546.25	12,486.31	5,546.25	12,486.31	5,546.25	12,486.31	5,546.25	12,486.31	5,546.25
▣ Art	27,118.79	6,527.79	27,118.79	6,527.79	27,118.79	6,527.79	27,118.79	6,527.79	27,118.79	6,527.79	27,118.79	6,527.79
▣ Envelopes	16,476.40	6,964.18	16,476.40	6,964.18	16,476.40	6,964.18	16,476.40	6,964.18	16,476.40	6,964.18	16,476.40	6,964.18
▣ Appliances	107,532.16	18,138.01	107,532.16	18,138.01	107,532.16	18,138.01	107,532.16	18,138.01	107,532.16	18,138.01	107,532.16	18,138.01
▣ Storage	223,843.61	21,278.83	223,843.61	21,278.83	223,843.61	21,278.83	223,843.61	21,278.83	223,843.61	21,278.83	223,843.61	21,278.83
▣ Binders	203,412.73	30,221.76	203,412.73	30,221.76	203,412.73	30,221.76	203,412.73	30,221.76	203,412.73	30,221.76	203,412.73	30,221.76
▣ Paper	78,479.21	34,053.57	78,479.21	34,053.57	78,479.21	34,053.57	78,479.21	34,053.57	78,479.21	34,053.57	78,479.21	34,053.57
▣ Technology	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95	836,154.03	145,454.95
▣ Machines	189,238.63	3,384.76	189,238.63	3,384.76	189,238.63	3,384.76	189,238.63	3,384.76	189,238.63	3,384.76	189,238.63	3,384.76
▣ Accessories	167,380.32	41,936.64	167,380.32	41,936.64	167,380.32	41,936.64	167,380.32	41,936.64	167,380.32	41,936.64	167,380.32	41,936.64
▣ Phones	330,007.06	44,515.73	330,007.06	44,515.73	330,007.06	44,515.73	330,007.06	44,515.73	330,007.06	44,515.73	330,007.06	44,515.73
▣ Copiers	149,528.03	55,617.82	149,528.03	55,617.82	149,528.03	55,617.82	149,528.03	55,617.82	149,528.03	55,617.82	149,528.03	55,617.82
Total	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02	2,297,200.86	286,397.02

Figure : Report 1 — Matrix expanded to Sub-Category level.

5.6 Report 2: Interactive Sales Analysis with Cascading Filters

The second Power BI report demonstrates cascading slicer functionality combined with multiple chart visuals to provide interactive geographic sales analysis. The report contains four visuals:

- Region slicer,
- State slicer
- horizontal bar chart showing Sales by Product Category
- clustered bar chart showing Sales Distribution by State.

Two slicers were configured on the report canvas. The first slicer displays the four Regions (Central, East, South, West) from the Region field in DimLocation. The second slicer displays all US States from the State field in DimLocation. Because both Region and State belong to the same DimLocation table which is related to FactSales, selecting a value in the Region slicer automatically filters the State slicer to show only states within that region. This cascading filter behaviour is achieved through the data model relationship between DimLocation and FactSales without requiring any DAX expressions or custom configuration.

The horizontal bar chart displays Sales by Product Category (Technology, Furniture, Office Supplies) and updates dynamically when either slicer is applied. The clustered bar chart displays Sales by State with Region colour coding, enabling geographic comparison. A Card visual displays the Total Sales figure which also updates with slicer selections.

The cascading filter demonstrates a key Power BI capability: when Region = West is selected, the State slicer immediately narrows to show only California, Colorado, Montana, Oregon, Utah, Washington, and other western states. Selecting a specific state then filters all visuals to that state's data, providing two levels of geographic filtering through a single interaction chain.

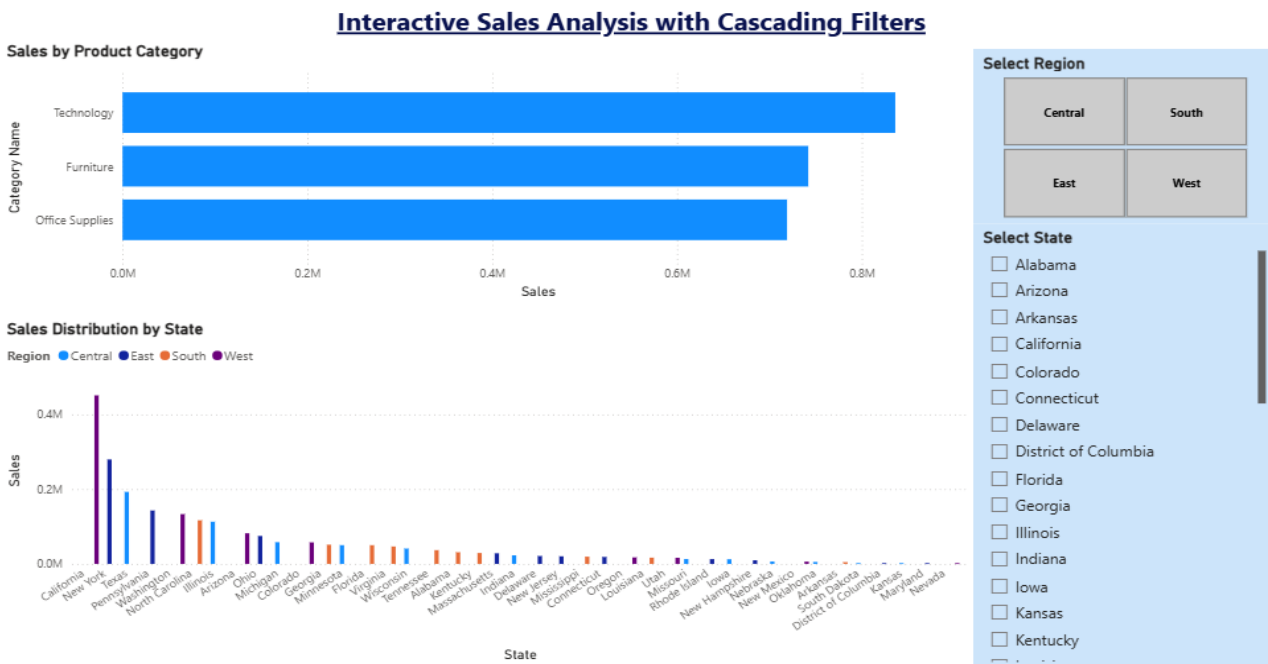


Figure : Report 2 — Initial state with no filters active. The Region slicer shows all four regions and the State slicer shows all US states. The bar chart displays total Sales across all categories and the clustered chart shows Sales by State for all regions.

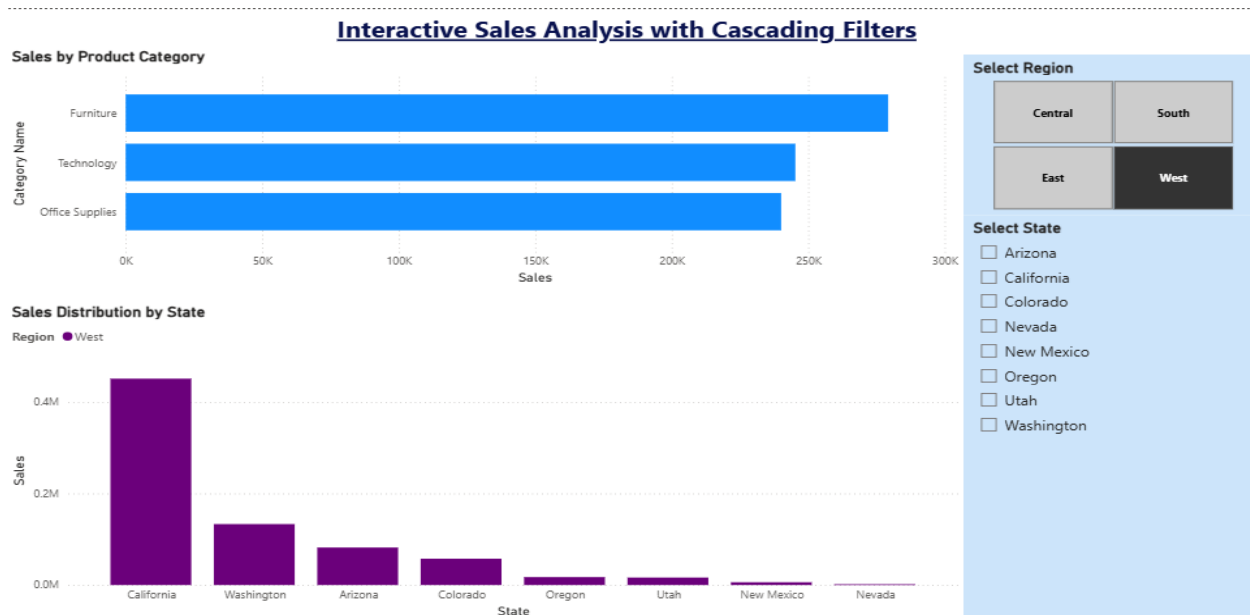


Figure : Report 2 — Cascading filter demonstrated. Selecting West in the Region slicer dynamically filters the State slicer to display only Western states. Both charts simultaneously update to show West region data only. Technology leads Sales in the West region.

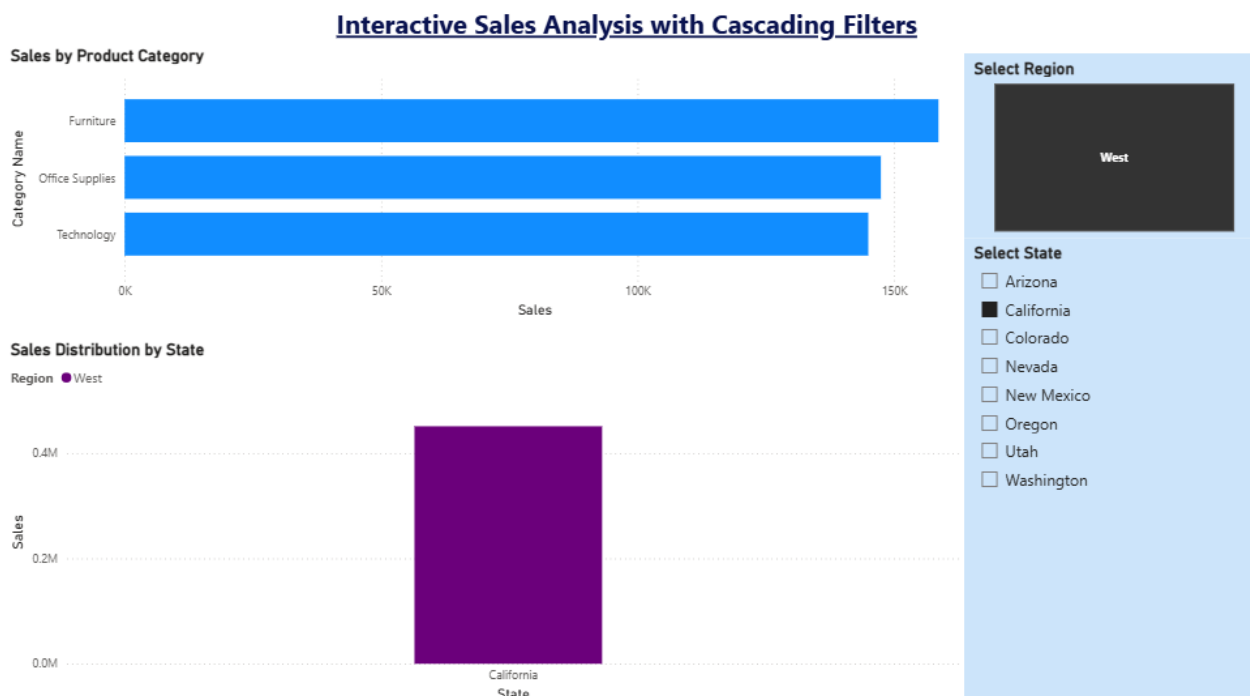


Figure : Report 2 — Second level cascade. With West already selected, clicking California in the State slicer filters all visuals to California-only data. The bar chart updates to show California category Sales and the clustered chart highlights California's position as the highest revenue state in the dataset.

5.7 Report 3: Sales and Profit Trend Analysis with Hierarchical Drill-Down

The third Power BI report demonstrates hierarchical drill-down functionality using the DateHierarchy created from the DimDate dimension table. The DateHierarchy follows the structure Year → Quarter → Month, enabling progressive time-based exploration of Sales and Profit data.

The report contains two visuals.

- The bar chart on the left displays total Sales by Year using the DateHierarchy on the X-axis.
- The line chart on the right displays total Profit by Year using the same DateHierarchy.

Both charts support independent drill-down navigation using the drill-down arrows located at the top right corner of each visual. Cross-filtering between the two charts was disabled so each chart drills down independently without affecting the other.

The drill-down navigation works as follows: at Year level the bar chart shows four bars for 2014 through 2017. Clicking the single down arrow navigates to Quarter level showing Q1 through Q4. Clicking again navigates to Month level showing individual monthly figures. The up arrow returns to the previous level. The line chart follows the same drill-down pattern independently.

Title: Sales and Profit Trend Analysis with Hierarchical Drill-Down

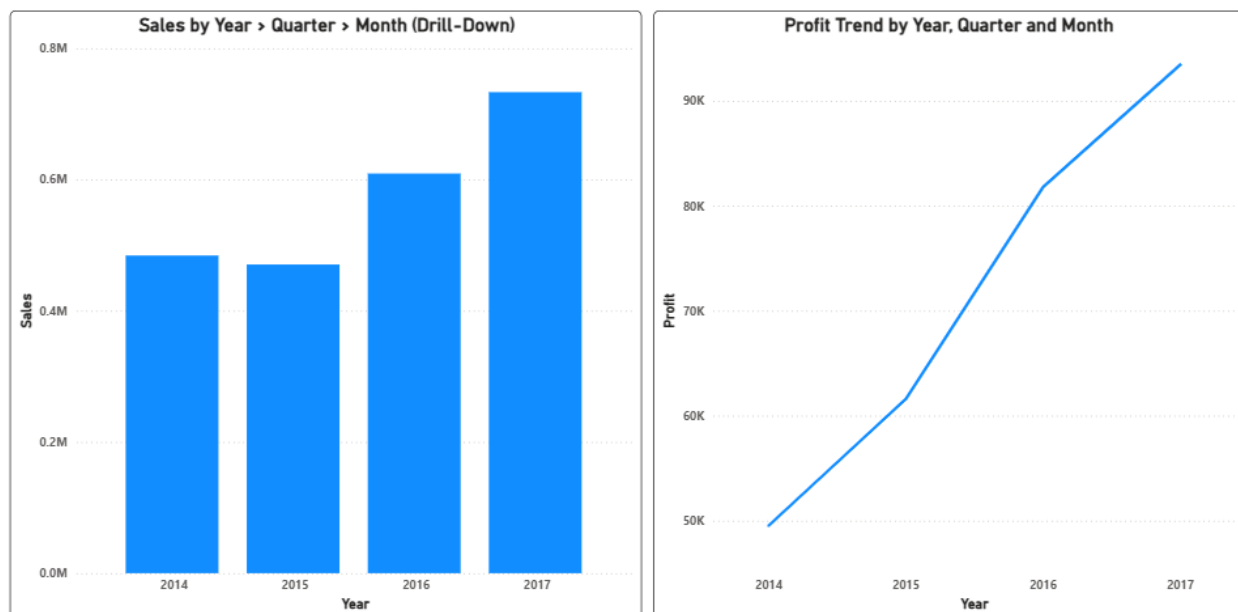


Figure Report 3 -Bar chart and line chart at Year level

Title: Sales and Profit Trend Analysis with Hierarchical Drill-Down

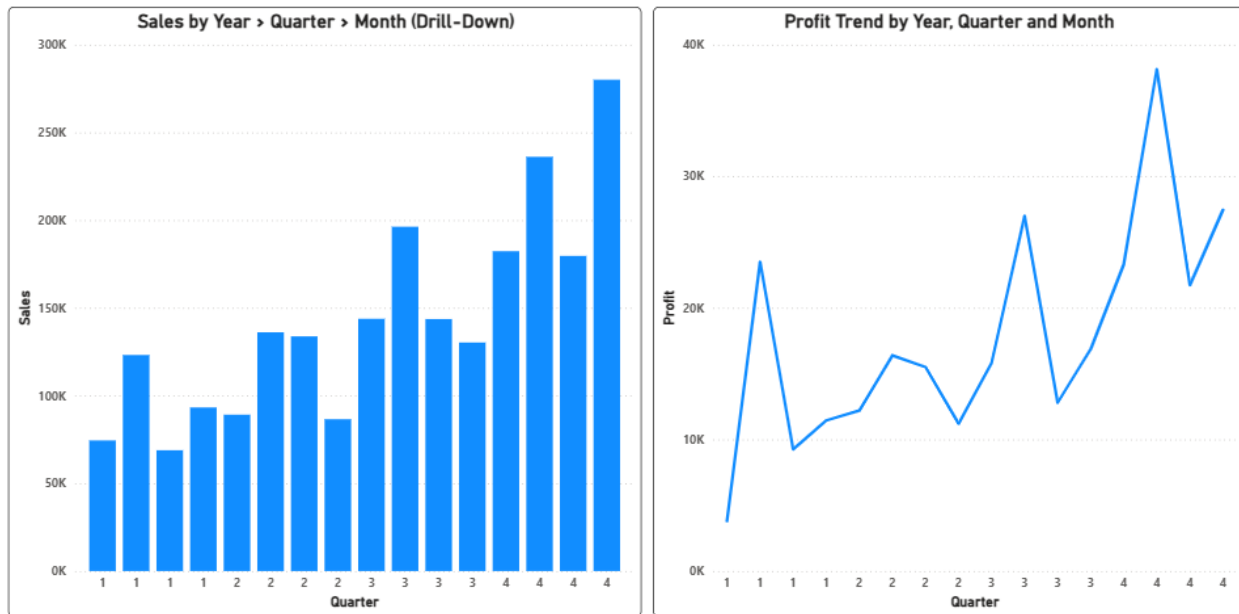


Figure : Report 3 — Sales bar chart and Profit line chart drilled down to Quarter level.

Title: Sales and Profit Trend Analysis with Hierarchical Drill-Down

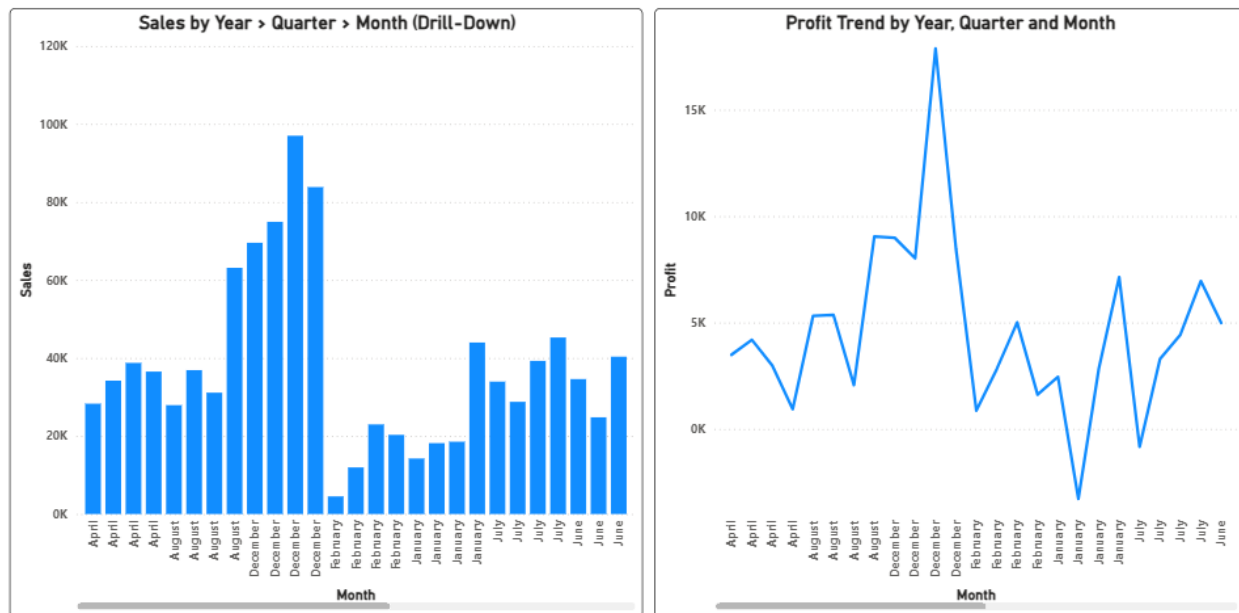


Figure : Report 3 — Sales bar chart and Profit line chart drilled to Month level

5.8 Report 4: Regional Sales Drill-Through Report

The fourth Power BI report demonstrates drill-through functionality, enabling users to right-click on a data point in a summary visual and navigate to a dedicated detail page filtered to that specific data point. This report consists of two pages: the main overview page and the Region Detail destination page.

The main page contains three visuals. A Treemap displays Total Sales by Region, with each region represented as a proportionally sized coloured rectangle (West, Central, East, South). A horizontal bar chart displays Sales by State, providing state-level detail within the regional overview. A Card visual shows Total Profit as a summary .

The Region Detail page was configured as the drill-through destination by placing the Region field from DimLocation in the Drill through field well of that page. This configuration means that right-clicking any region in the Treemap on the main page presents a context menu with the drill-through option. Selecting it navigates to the Region Detail page, which automatically filters all its visuals to show data for only the selected region.

The Region Detail page contains four visuals: two Card visuals showing Total Sales for Selected Region and Total Profit for Selected Region, a pie chart showing Sales by Ship Mode, a horizontal bar chart showing Sales by Category, and a table showing Customer Orders Detail with CustomerName, Segment, Sales, and Profit columns. A back arrow button appears automatically in the top-left corner of the detail page, allowing users to return to the main page. This drill-through design enables a two-level analytical journey: from a high-level regional sales overview to granular customer and category detail for any selected region, all within a single report file.

Regional Sales Overview — Right-Click a Region to Drill Through

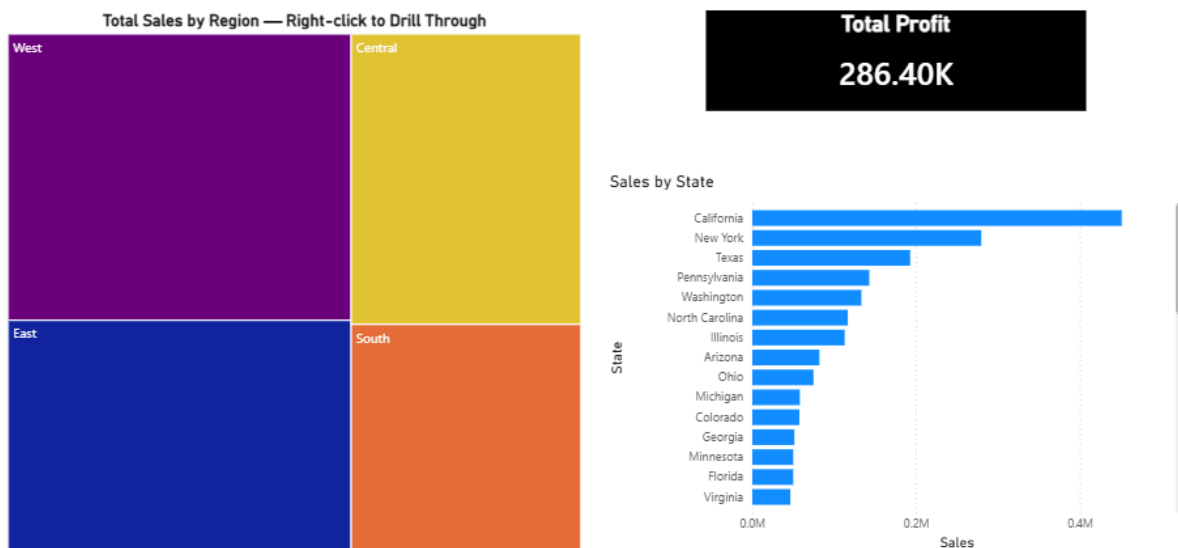


Figure : Report 4 — Main page showing Treemap of Total Sales by Region, Sales by State bar chart, and Total Profit card.

Regional Sales Overview — Right-Click a Region to Drill Through

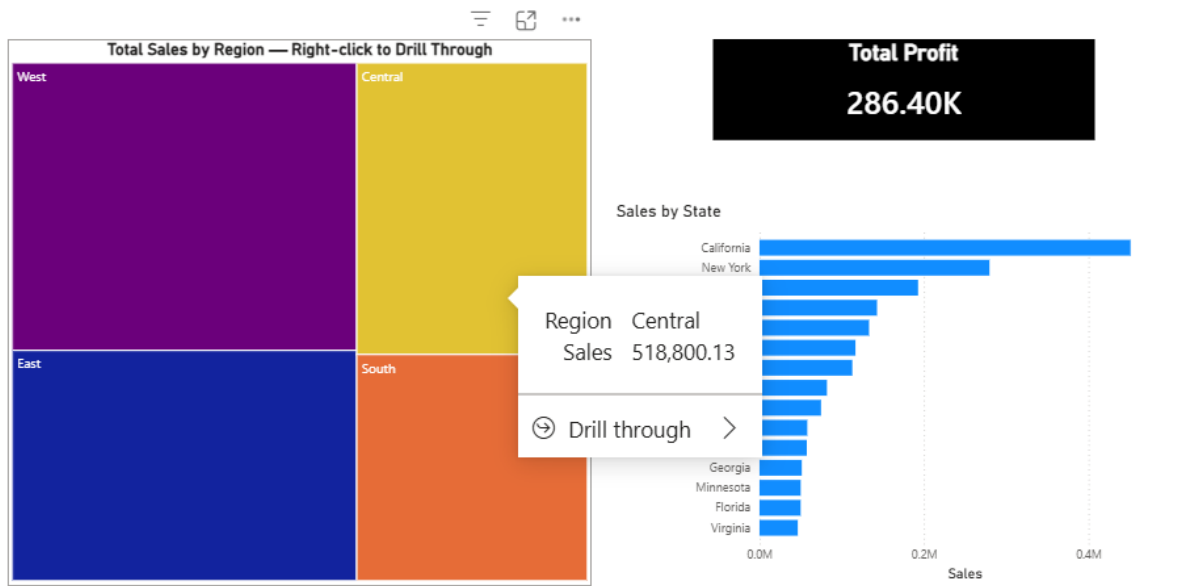


Figure : Report 4 — Right-clicking the Central region in the Treemap reveals the context menu with Drill through → Report 4 - Region Detail option. This demonstrates the drill-through entry point configured through the Region field in the Drill through field well of the destination page.



Region Detail — Sales and Customer Analysis

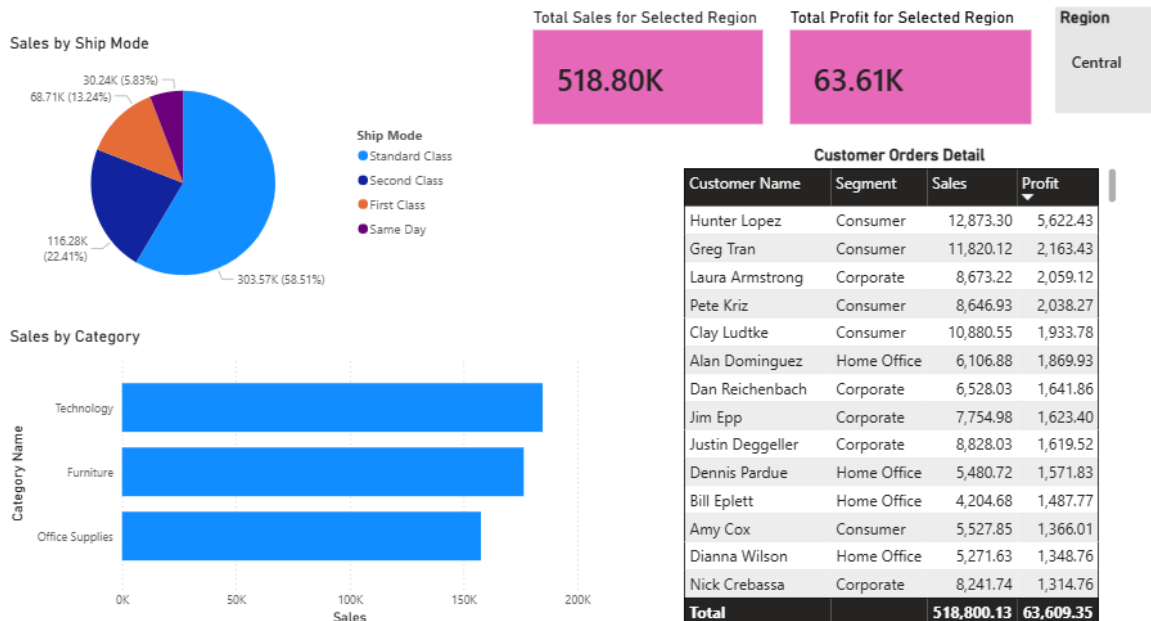


Figure : Report 4 — Region Detail page after drilling through from Central region. All visuals are automatically filtered to Central region data. Total Sales for Central region and Total Profit are shown. The pie chart shows Standard Class as the dominant ship mode. The Customer Orders table lists individual customer transactions within the Central region.

5.9 Publishing to Power BI Service

All four reports were published to Power BI Service for online access:

1. In Power BI Desktop, click File → Publish → Publish to Power BI.
2. Select the destination workspace (e.g., My Workspace).
3. After publishing, navigate to app.powerbi.com to verify all four report pages load correctly.
4. Open each report page and confirm that slicers, drill-down, and drill-through interactions function as expected in the online environment.

Category Name	2014		2015		2016		2017		2018		Total	
	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit	Sales	Profit
ire	741,999.80	18,451.27	741,999.80	18,451.27							8,451.27	
s	206,965.53	-17,725.48	206,965.53	-17,725.48							7,725.48	
cases	114,880.00	-3,472.56	114,880.00	-3,472.56							3,472.56	
things	91,705.16	13,059.14	91,705.16	13,059.14							3,059.14	
s	326,449.10	26,590.17	326,449.10	26,590.17							6,590.17	
Supplies	719,047.03	122,490.80	719,047.03	122,490.80							2,490.80	
ies	46,673.54	-1,189.10	46,673.54	-1,189.10							1,189.10	
ners	3,024.28	949.52	3,024.28	949.52							949.52	
s	12,486.31	5,546.25	12,486.31	5,546.25							5,546.25	
	27,118.79	6,527.79	27,118.79	6,527.79							6,527.79	
opes	16,476.40	6,964.18	16,476.40	6,964.18							6,964.18	
ances	107,532.16	18,138.01	107,532.16	18,138.01							8,138.01	
ge	223,843.61	21,278.83	223,843.61	21,278.83							1,278.83	
rs	203,412.73	30,221.76	203,412.73	30,221.76							0,221.76	
	78,479.21	34,053.57	78,479.21	34,053.57							4,053.57	
logy	836,154.03	145,454.95	836,154.03	145,454.95							5,454.95	
ines	189,238.63	3,384.76	189,238.63	3,384.76							3,384.76	
ories	167,380.32	41,936.64	167,380.32	41,936.64							167,380.32	41,936.64
es	330,007.05	44,515.73	330,007.05	44,515.73							330,007.05	44,515.73
rs	149,528.03	55,617.82	149,528.03	55,617.82							149,528.03	55,617.82
	2,297,200.86	286,397.02	2,297,200.86	286,397.02							2,297,200.86	286,397.02

Figure : Reports published and accessible in Power BI Service (online)

While the publication to Power BI Service completed successfully, the reports appear blank in the online environment with a data source access error. This is because the reports use a live connection to a locally hosted SSAS cube, which requires an on-premises data gateway with configured Windows authentication to serve data to the cloud. All four reports display complete and correct data when accessed through Power BI Desktop, as evidenced by the screenshots provided in the sections above.

6. Summary

This report documented the complete implementation of a Business Intelligence solution for the Superstore Retail Sales Data Warehouse as part of IT3021 Assignment 2.

The SSAS Multidimensional Cube (Cube_Superstore) was successfully built upon the Superstore_DW star schema. The cube includes four measures (Sales, Quantity, Profit, Discount), five connected dimensions (Date, Customer, Product, Location, ShipMode), and three user-defined hierarchies (DateHierarchy, LocationHierarchy, ProductHierarchy). Multiple processing challenges related to duplicate attribute keys were identified and resolved through correct composite key configuration and attribute relationship design.

All five standard OLAP operations were demonstrated using Excel PivotTables connected live to the cube: Roll-Up and Drill-Down via the DateHierarchy, Slice via a single Region filter, Dice via combined Region and Year filters, and Pivot by rotating Category from Rows to Columns.

Four Power BI reports were developed using a live cube connection and published to Power BI Service: a Matrix report with product and year groupings, a cascading slicer report for geographic analysis, a drill-down report on date hierarchy, and a drill-through report for region-level detail navigation.

The complete solution demonstrates the integration of SQL Server Analysis Services, Excel OLAP analysis, and Power BI interactive reporting to deliver a robust and flexible BI platform for Superstore retail analytics.